CI/CD for Node.js Retail App

Course Name: Devops

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number:DO-01

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Academic Year:2022-2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features
2. Overall Architecture / Workflow
3. Tools & Technologies Used (*If applicable)*

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Conclusion

Future Scope & Enhancements

1. Problem Statement & Objectives

1.1. Problem Statement

Manual software deployment in retail environments leads to "Human Error," inconsistent environments, and slow release cycles. A developer might forget to run tests or use a different version of Node.js than the server, causing production outages.

1.2. Project Objectives

- To eliminate manual deployment steps through **Automation**.
- To ensure high code quality by enforcing **Automated Unit Testing**.
- To achieve environment consistency using **Docker Containerization**.

1.3. Scope of the Project

The scope covers the CI/CD pipeline from the moment code is pushed to GitHub, through the testing phase, ending with the creation of a verified Docker image.

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2. Proposed Solution

2.1. Key Features

- **Trigger-based Automation:** Automatic execution on every code push.
- **Quality Gates:** Tests must pass for the build to proceed.
- **Immutable Artifacts:** Every build creates a unique Docker container.
- **Failure Notifications:** Immediate feedback to developers if a build fails.

2.2. Overall Architecture / Workflow

The workflow follows a linear "Commit-Test-Build" logic.

2.3. Tools & Technologies Used

- **Language:** Node.js (Runtime)
- **Version Control:** GitHub
- **CI/CD Engine:** GitHub Actions
- **Containerization:** Docker
- **Environment:** GitHub Codespaces (Development)

3. Results & Output

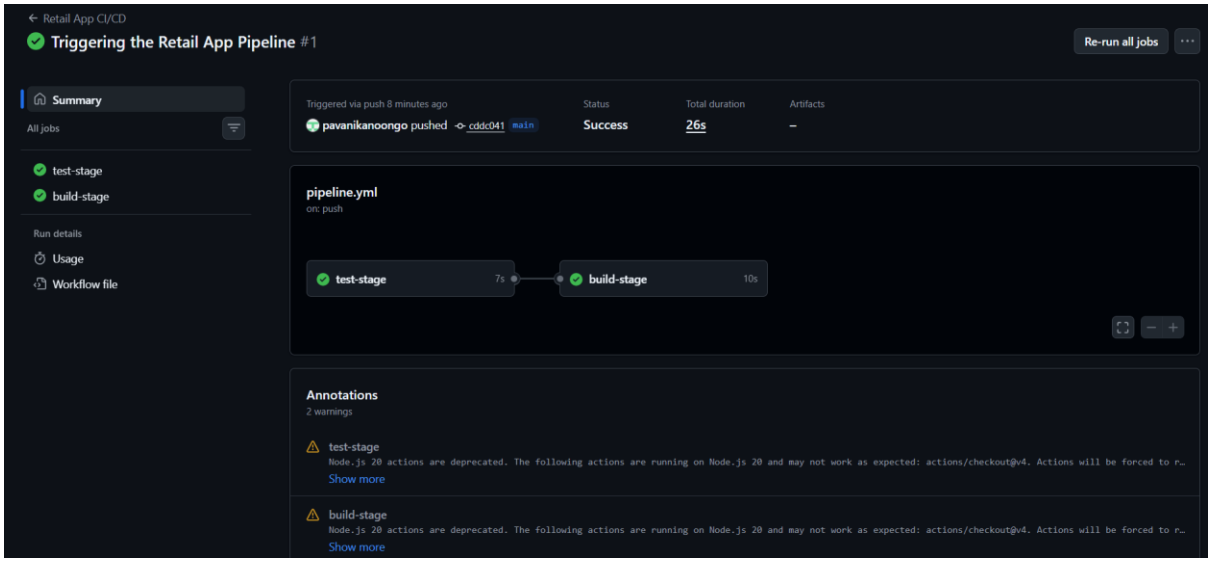
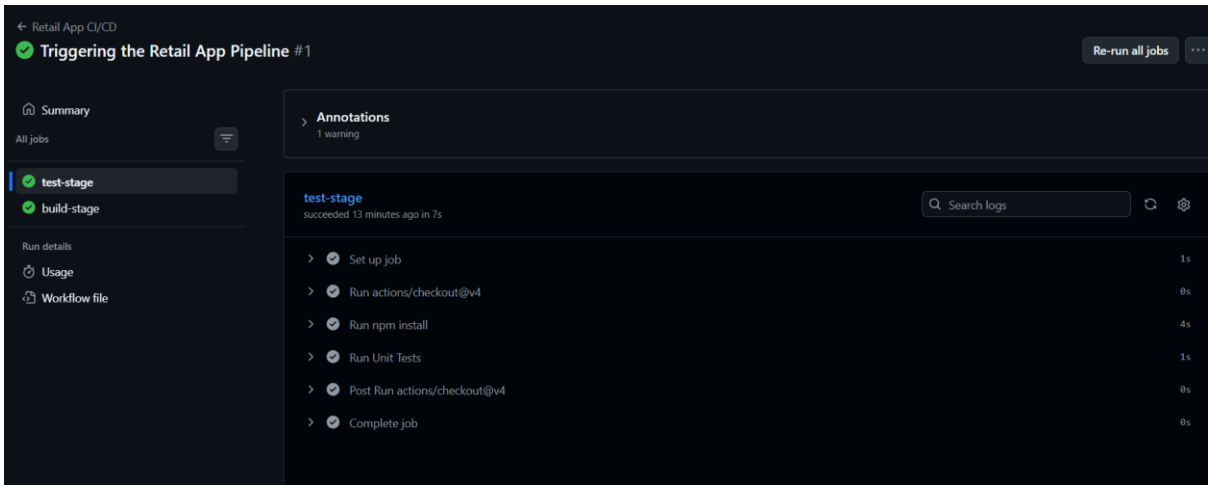
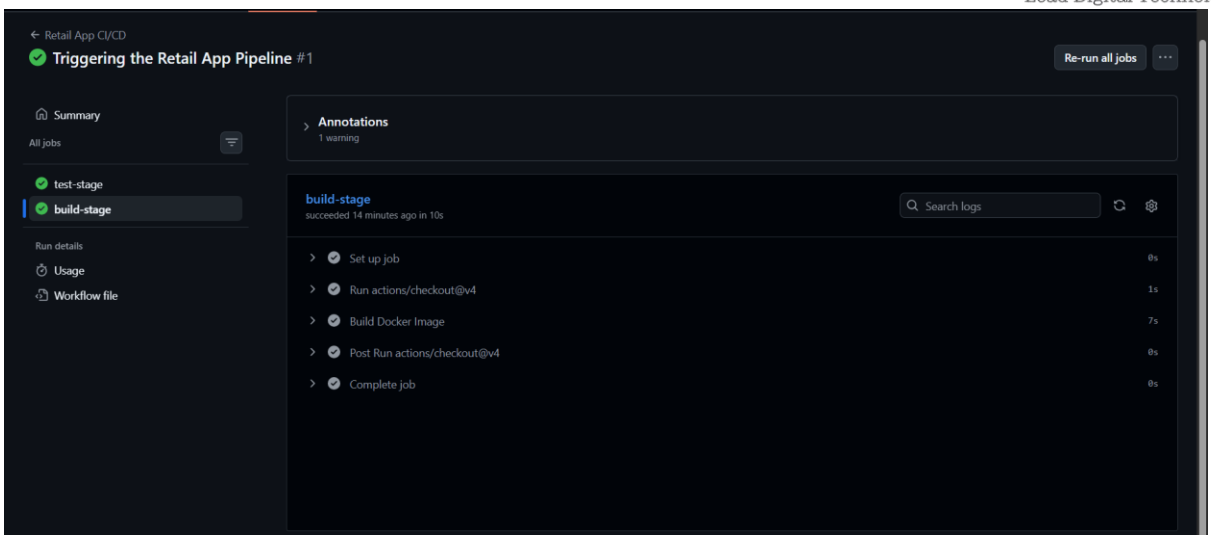
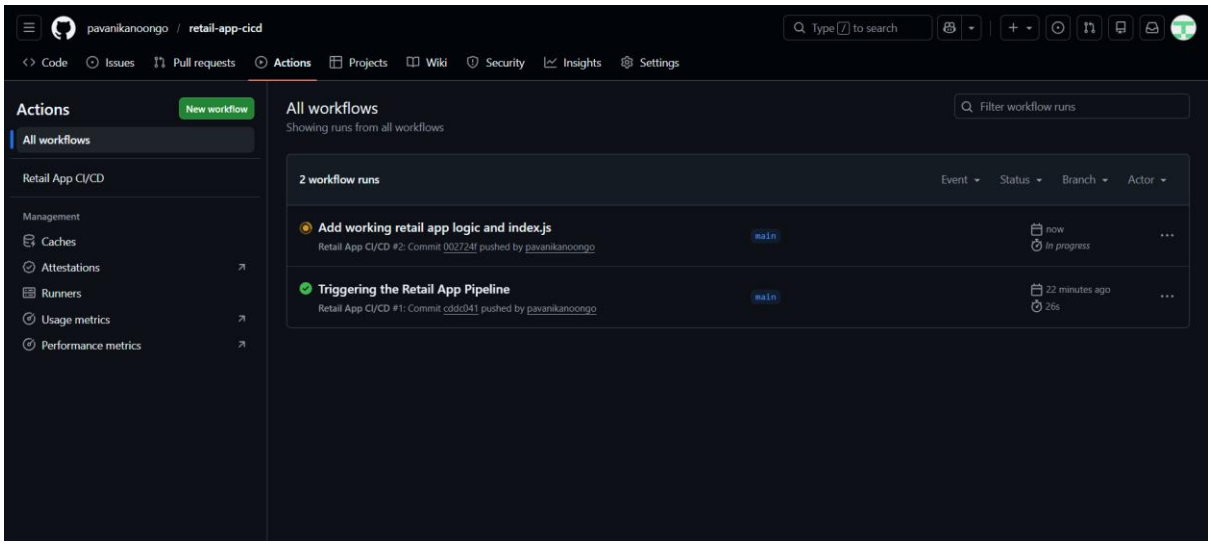
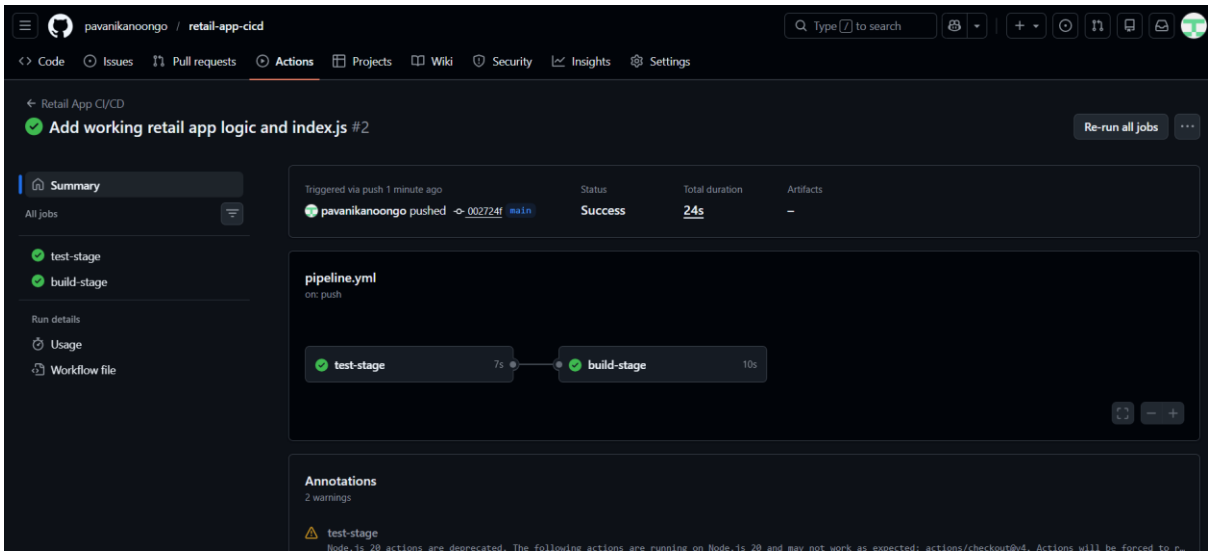


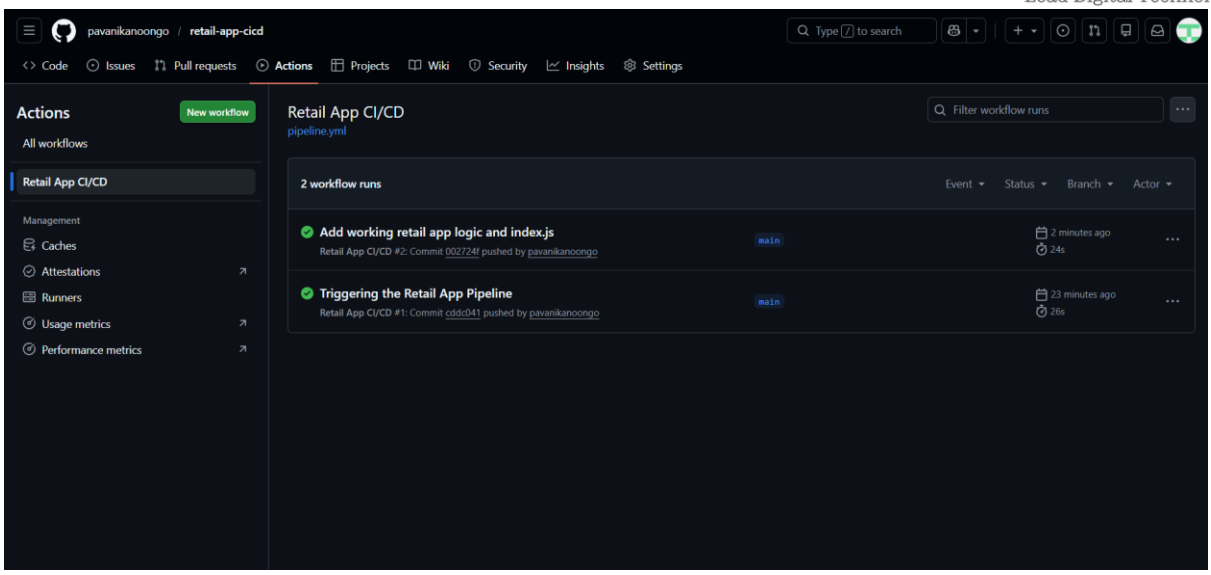
Fig.1 Automated CI/CD Workflow Execution




Retail App CI/CD
Triggering the Retail App Pipeline #1
Re-run all jobs
Summary
All jobs
test-stage
build-stage
Run details
Usage
Workflow file
Annotations
1 warning
build-stage
succeeded 14 minutes ago in 10s
Search logs
Set up job
Run actions/checkout@v4
Build Docker Image
Post Run actions/checkout@v4
Complete job


pavanikanoongo / retail-app-cicd
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Actions
New workflow
All workflows
Retail App CI/CD
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
All workflows
Showing runs from all workflows
Filter workflow runs
2 workflow runs
Event
Status
Branch
Actor
Add working retail app logic and index.js
Retail App CI/CD #2: Commit 002724f pushed by pavanikanoongo
main
now
In progress
Triggering the Retail App Pipeline
Retail App CI/CD #1: Commit cddc041 pushed by pavanikanoongo
main
22 minutes ago
26s


pavanikanoongo / retail-app-cicd
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Retail App CI/CD
Add working retail app logic and index.js #2
Re-run all jobs
Summary
All jobs
test-stage
build-stage
Run details
Usage
Workflow file
Triggered via push 1 minute ago
pavanikanoongo pushed 002724f main
Status
Success
Total duration
24s
Artifacts
–
pipeline.yml
on: push
test-stage
7s
build-stage
10s
Annotations
2 warnings
test-stage



4. Conclusion

This project successfully demonstrates a modern DevOps approach to software delivery. By integrating GitHub Actions and Docker, we transformed a standard Node.js application into a portable, enterprise-ready service.

Key Learning: The most critical part of a professional environment isn't just the code itself, but the **automated gates** that ensure only high-quality, tested code reaches the end user.

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5. Future Scope & Enhancements

- **SAST Integration:** Adding security scanning (Snyk or npm audit) directly into the pipeline.
- **Auto-Deployment:** Extending the pipeline to deploy the Docker image automatically to a Kubernetes cluster.
- **Monitoring:** Integrating Prometheus/Grafana to track the health of the retail app after deployment.